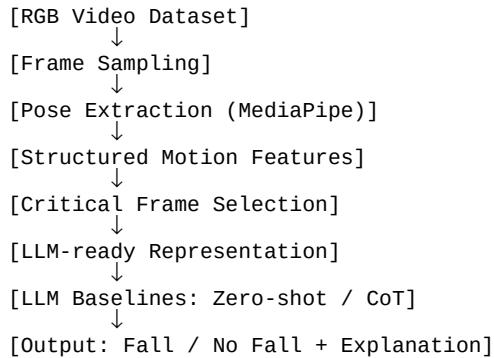


GenAI-based Fall Detection Project

1. Project Architecture



2. Dataset Options (with Links)

- 1 Le2i Fall Detection Dataset: https://github.com/YifeiYang210/Fall_Detection_dataset
- 2 UBFC Fall Detection Dataset: https://search-data.ubfc.fr/FR-13002091000019-2024-04-09_Fall-Detection-Dataset.html
- 3 URFD Dataset: <https://fenix.ur.edu.pl/~mkepski/ds/uf.html>
- 4 NTU RGB+D Dataset: <https://rose1.ntu.edu.sg/dataset/actionRecognition/>
- 5 MD4FD Meta Dataset: <https://github.com/ispstiima/MD4FD-Dataset>

3. Detailed Steps

- 1 Split dataset into train/validation/test at VIDEO level (avoid leakage)
- 2 Sample frames from videos (e.g., 2 FPS)
- 3 Extract pose landmarks using MediaPipe
- 4 Compute motion features (torso angle, velocity, positions)
- 5 Select critical frames (before, during, after event)
- 6 Convert features into structured JSON/text format
- 7 Run zero-shot LLM baseline
- 8 Run Chain-of-Thought (CoT) reasoning baseline
- 9 Evaluate using accuracy, precision, recall, F1 score